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1 Network Configurations in Microsoft Fabric

1.1 Overview

Microsoft Fabric is a unified SaaS platform for analytics and AI, bringing together Data Factory, Data Engineering, Data Science, Data Warehousing, Real-Time Intelligence and Power BI around a single data lake: **OneLake**.

Network security in Fabric is built around three fundamental pillars:

Pillar	Objective	Key Features
Inbound Protection	Control who accesses Fabric	Conditional Access, Private Links (tenant/workspace), IP Firewall
Secure Outbound Access	Connect Fabric to protected data sources	Trusted Workspace Access, Managed Private Endpoints, VNet/On-prem Gateways

Pillar	Objective	Key Features
Outbound Protection	Prevent data exfiltration	Outbound Access Policies, allowed destination rules

Combining **Inbound + Outbound Protection** provides complete **Data Exfiltration Protection (DEP)**.

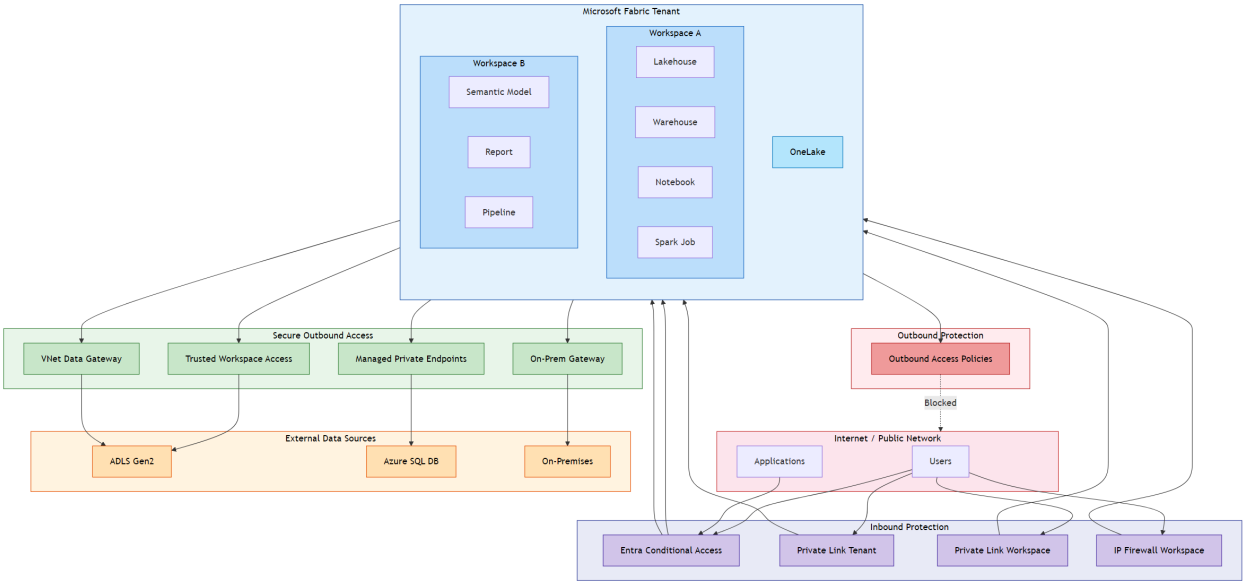


Figure 1: Diagram 1

1.2 Secure by Default

Microsoft Fabric is **secure by default** without any additional configuration:

- **Authentication**: Every interaction is authenticated via **Microsoft Entra ID**
- **Encryption in transit**: All traffic is encrypted using **TLS 1.2** minimum (TLS 1.3 negotiation when available)
- **Encryption at rest**: All data in OneLake is automatically encrypted
- **Microsoft backbone network**: Internal communications between Fabric experiences travel through the Microsoft private network, never over the public internet
- **Secure endpoints**: The Fabric backend is protected by a virtual network and is not directly accessible from the public internet

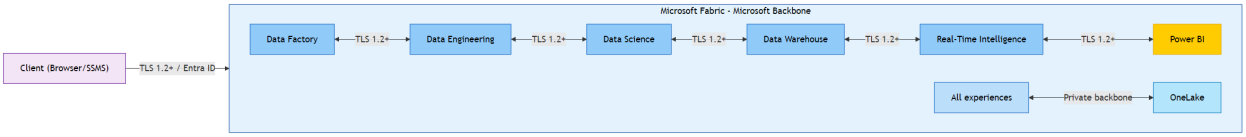


Figure 2: Diagram 2

1.3 Inbound Protection

Inbound protection controls traffic entering Fabric from the internet or corporate networks.

1.3.1 Entra Conditional Access

Microsoft Entra ID Conditional Access is the **Zero Trust** approach for securing access to Fabric. Access decisions are made at authentication time by evaluating contextual signals.

Evaluated signals:

Signal	Policy examples
Users and groups	Target specific populations
Location / IP	Allow only certain IP ranges or countries
Devices	Require compliant devices (Intune)
Applications	Apply rules per Fabric application
Sign-in risk	Block high-risk sign-ins

Possible decisions: Block, Grant, Require MFA, Require compliant device.

Prerequisites: Microsoft Entra ID P1 license (often included in Microsoft 365 E3/E5).

> **Note:** Conditional Access policies apply to Fabric and related Azure services (Power BI, Azure Data Explorer, Azure SQL Database, Azure Storage). This may be considered too broad for some scenarios.

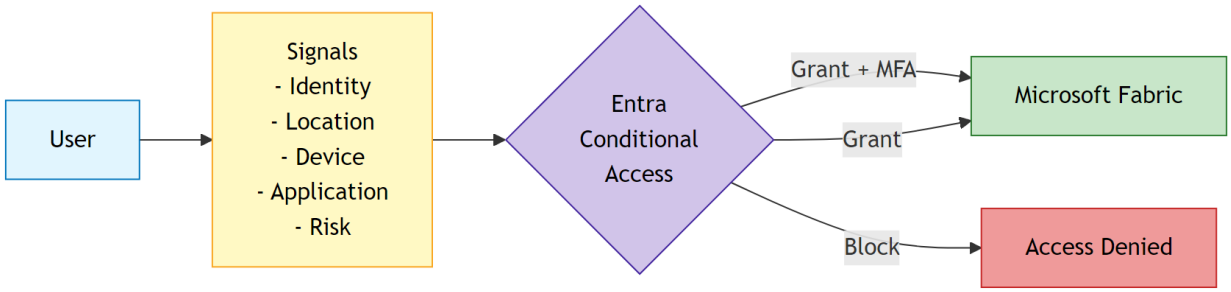


Figure 3: Diagram 3

1.3.2 Tenant-Level Private Link

Tenant-level Private Link provides **perimeter-based protection** for the entire Fabric tenant.

Concept: An Azure Private Endpoint is created in the customer's VNet, establishing a private tunnel to Fabric via the Microsoft backbone. Fabric becomes inaccessible from the public internet.

Two key settings:

Setting	Effect
Azure Private Links (enabled)	Traffic from the VNet to supported endpoints goes through the Private Link
Block Public Internet Access (enabled)	Fabric is no longer accessible from the public internet; only Private Link is allowed

Configuration scenarios:

Private Link	Block Public Access	Behavior
Yes	Yes	Access only via Private Endpoint. Unsupported endpoints blocked.
Yes	No	VNet traffic via Private Link. Public internet traffic allowed.
No	-	Standard access via public internet.

Considerations:

- **All users** must connect through the private network (VPN, ExpressRoute)
- **Bandwidth** impact: static resources (CSS, images) also transit through the Private Endpoint
- Some features have **limitations** (Publish to Web disabled, Copilot not supported, Power BI PDF/PowerPoint export disabled)
- **Spark Starter Pools** are disabled (replaced by custom pools in the managed VNet)

1.3.3 Workspace-Level Private Link

Workspace-level Private Link offers **granular** control: specific workspaces are protected by Private Link while others remain publicly accessible.

Key characteristics:

- **1:1** relationship between a workspace and its Private Link Service
- A Private Link Service can have **multiple Private Endpoints** (from different VNets)
- A VNet can connect to **multiple workspaces** via separate Private Endpoints
- Public access can be restricted **independently** per workspace
- Public workspaces can be secured with Entra Conditional Access or IP Filtering

Supported items (Public Preview August 2025, GA September 2025): Lakehouse, Shortcut, Notebook, ML Experiment/Model, Pipeline, Warehouse, Dataflows, Eventstream, Mirrored DB

Roadmap: Support for Power BI, Databases, Data Activator and more.

> **Key difference from Tenant-Level Private Link:** Workspace-level Private Link allows you to protect only sensitive workspaces without impacting the entire tenant. This is

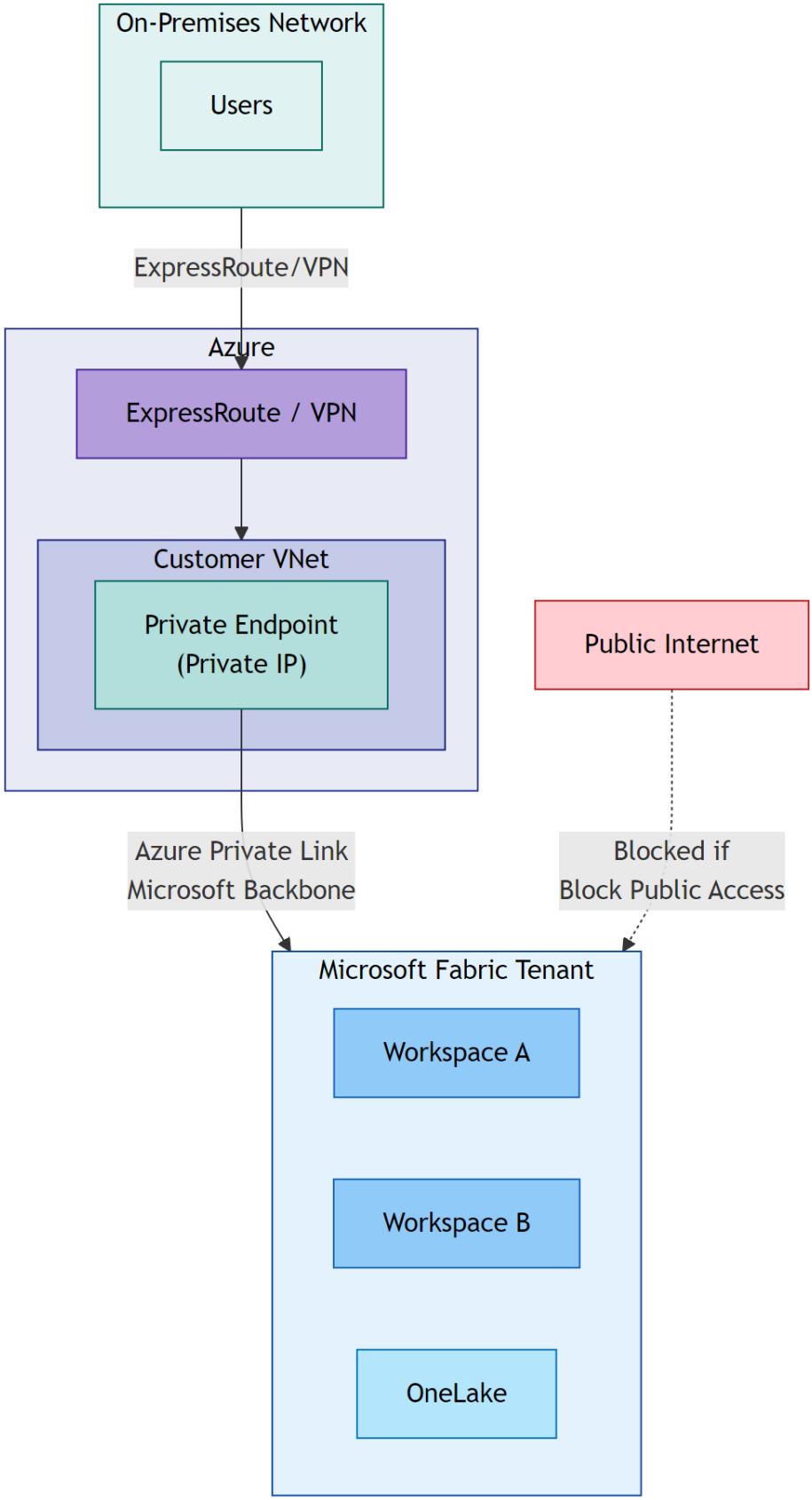


Figure 4: Diagram 4

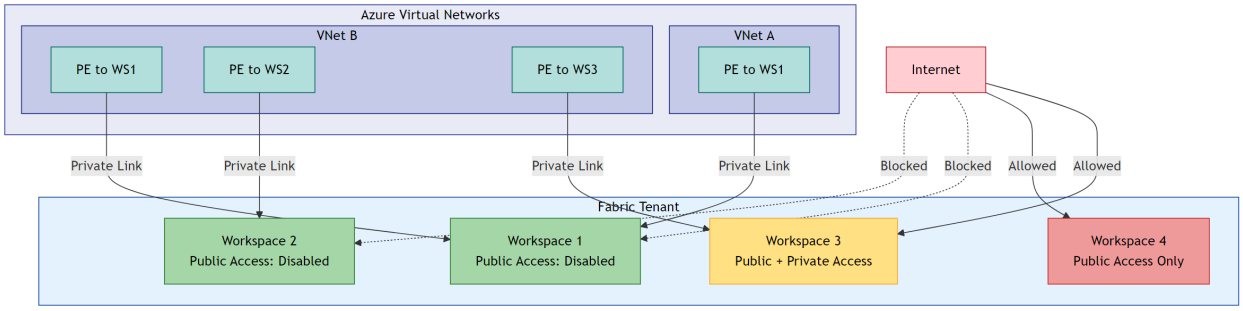


Figure 5: Diagram 5

the recommended approach for organizations that cannot force all users onto a private network.

1.3.4 Workspace IP Firewall

The workspace IP firewall is the simplest solution to restrict access to a workspace from specific public IP ranges.

Concept: Only explicitly allowed IP addresses or IP ranges can access the workspace. No Azure network infrastructure is required.

Supported items (Public Preview October 2025, GA Q1 2026): Lakehouse, Shortcut, Notebook, ML Experiment/Model, Pipeline, Warehouse, Dataflows, Eventstream, Mirrored DB

Roadmap: Support for Power BI, Databases, Data Activator.

1.3.5 Inbound Options Comparison

Criteria	Conditional Access	Private Link Tenant	Private Link Workspace	IP Firewall
Granularity	Tenant (per policy)	Tenant	Workspace	Workspace
Azure infrastructure required	No	VNet + PE	VNet + PE per WS	No
Complexity	Low	High	Medium	Low
Approach	Zero Trust (identity)	Perimeter (network)	Perimeter (network)	IP-based
User impact	Transparent (MFA)	VPN/ER mandatory	VPN/ER for protected WS	None if IP allowed
Additional cost	Entra ID P1	VNet + PE + ER/VPN	VNet + PE	None
Status	GA	GA	GA (Sept 2025)	GA (Q1 2026)

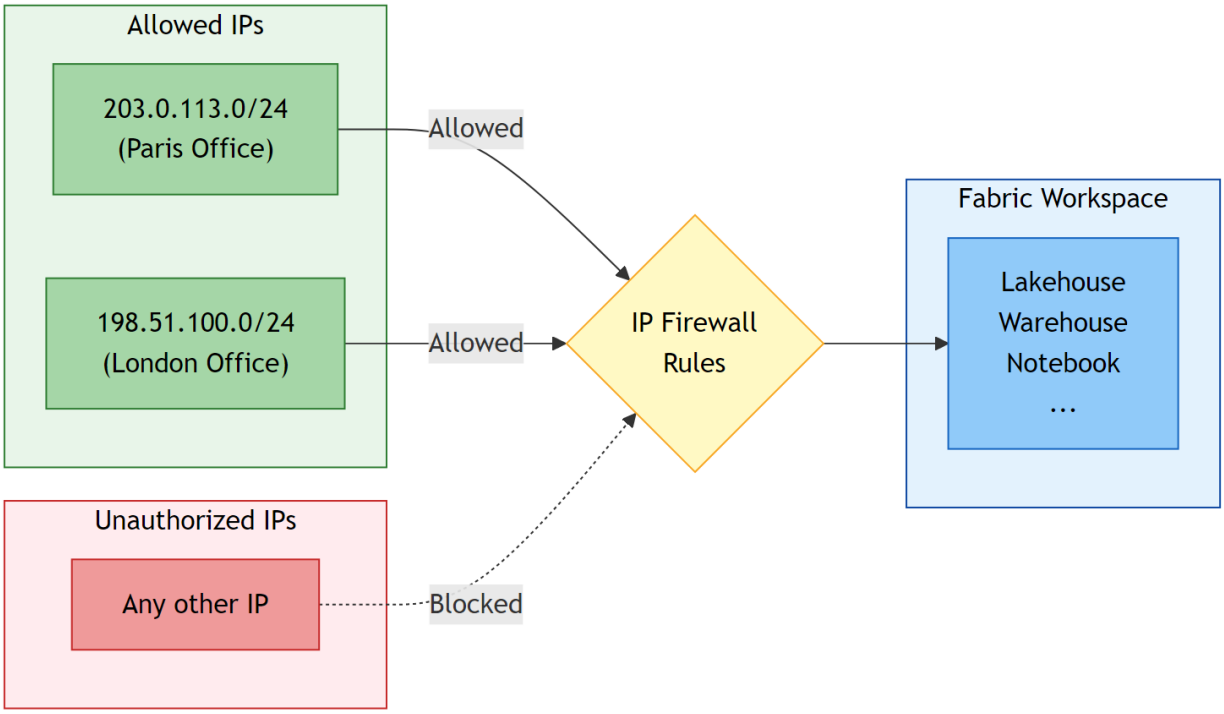


Figure 6: Diagram 6

1.4 Secure Outbound Access

Secure outbound access allows Fabric to connect to data sources protected by firewalls or private networks.

1.4.1 Trusted Workspace Access

Trusted Workspace Access allows specific Fabric workspaces to securely access **firewall-enabled ADLS Gen2 accounts**.

Concept: The Fabric workspace has a **workspace identity** (managed identity). **Resource Instance Rules** are configured on the storage account to authorize only specified workspaces.

Prerequisites: - Workspace associated with a **Fabric F SKU** capacity (not supported on Trial) - **Workspace Identity** created and configured as Contributor - Authentication principal with Azure RBAC role on the storage account (Storage Blob Data Contributor/Owner/Reader) - Resource Instance Rule configured via **ARM Template** or **PowerShell**

Supported scenarios:

Method	Description
OneLake Shortcut	ADLS Gen2 shortcut in a Lakehouse
Pipeline	Data copy from firewall-enabled ADLS Gen2
T-SQL COPY INTO	Ingestion into a Warehouse
Semantic Model (import)	Model connected to ADLS Gen2

Method	Description
AzCopy	High-performance load to OneLake

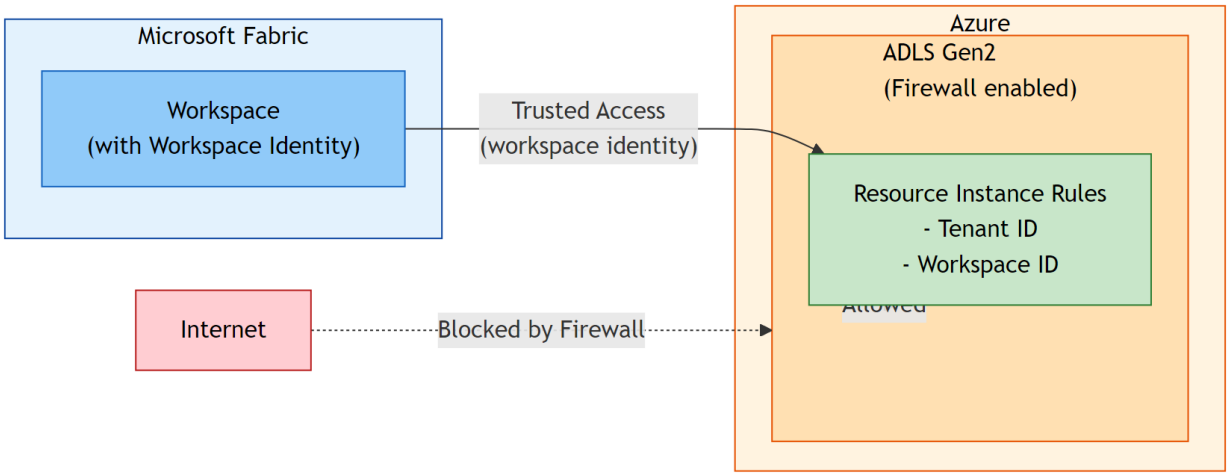


Figure 7: Diagram 7

1.4.2 Managed Private Endpoints

Managed Private Endpoints enable secure, private connections to Azure data sources from Fabric workloads without exposing them to the public network.

Concept: Microsoft Fabric creates and manages Private Endpoints in a **managed VNet** dedicated to the workspace. Workspace admins specify the resource ID of the source, the target sub-resource, and a justification.

Supported sources: Azure Storage, Azure SQL Database, Azure Cosmos DB, Azure Key Vault, and many more.

Supported items: - Data Engineering (Spark/Python Notebooks, Lakehouses, Spark Job Definitions) - Eventstream (Preview)

Prerequisites: - Supported on Fabric Trial and all F SKU capacities - Data Engineering workload must be available in both the tenant region AND the capacity region

Limitations: - OneLake shortcuts do not yet support ADLS Gen2 / Blob Storage connections via MPE - Cross-region workspace migration not supported - FQDN-based creation (Private Link Service) only via REST API

1.4.3 Managed Virtual Networks

Managed VNets are virtual networks created and managed by Fabric for each workspace. They provide network isolation for Spark workloads.

Automatically provisioned when: 1. **Managed Private Endpoints** are added to the workspace 2. The **Private Link** setting is enabled and a Spark job is executed

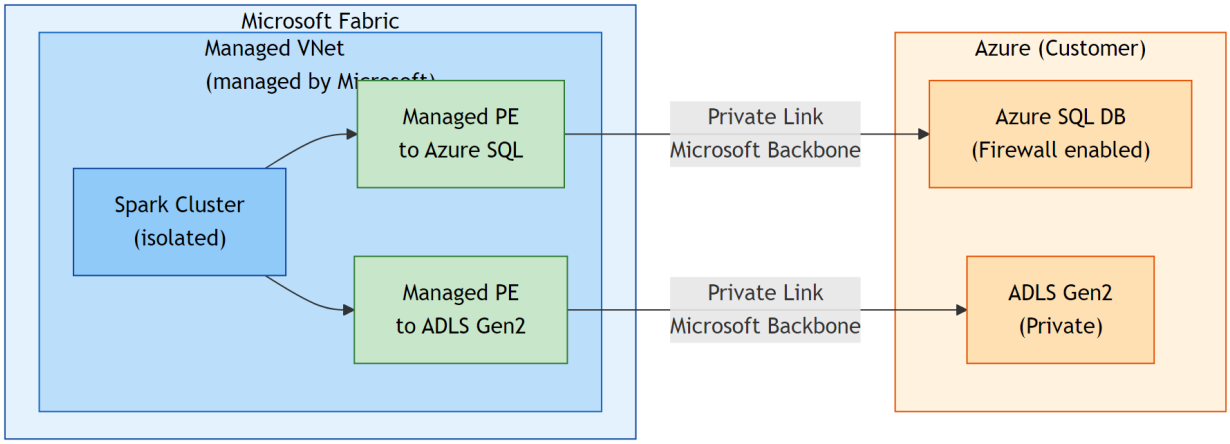


Figure 8: Diagram 8

Characteristics: - Complete Spark cluster isolation (dedicated network) - No need to size subnets (managed by Fabric) - **Starter Pools** are disabled (on-demand Custom Pools, 3-5 min startup) - Not supported in Switzerland West and West Central US regions

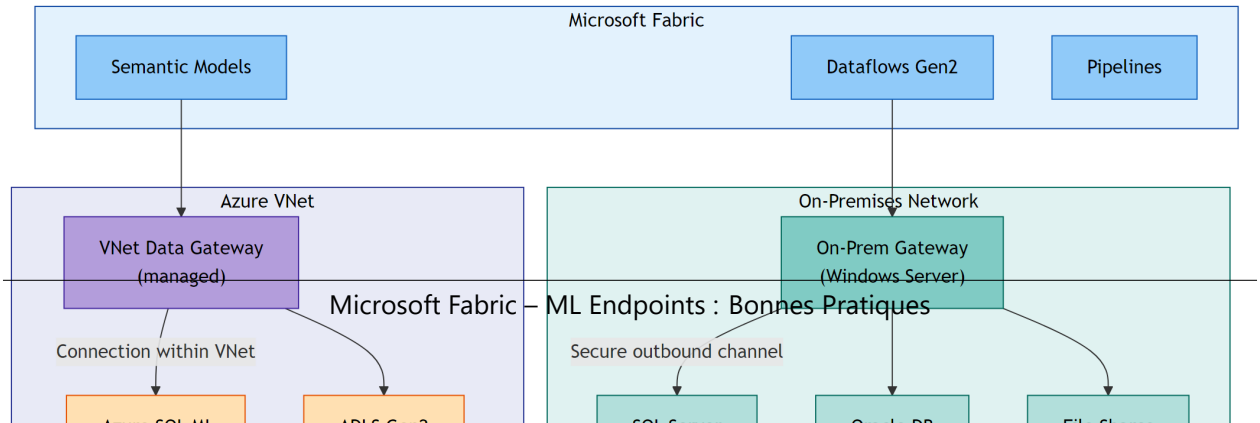
1.4.4 Data Gateways

1.4.4.1 On-Premises Data Gateway A gateway installed on a server within the corporate network, acting as a bridge between on-premises data sources and Fabric.

Characteristic	Detail
Installation	Windows server in the internal network
Protocol	Secure outbound channel (no inbound port opening)
Sources	Any source accessible from the gateway server
Management	Manual (updates, high availability)

1.4.4.2 VNet Data Gateway A managed gateway deployed into a customer's Azure VNet, enabling connections to Azure services within the VNet without an on-premises gateway.

Characteristic	Detail
Deployment	Injection into an existing Azure VNet
Management	Managed by Microsoft
Sources	Azure services in the VNet or peered VNets
Workloads	Dataflows Gen2, Semantic Models



Tag	Service	Direction	Regional
`Power BI`	Power BI and Microsoft Fabric	Inbound/Outbound	Yes
`DataFactory`	Azure Data Factory	Inbound/Outbound	Yes
`DataFactoryManagement`	On-premises pipeline	Outbound	No
`EventHub`	Azure Event Hubs	Outbound	Yes
`KustoAnalytics`	Real-Time Analytics	Inbound/Outbound	No
`SQL`	Warehouse	Outbound	Yes
`PowerQueryOnline`	Power Query Online	Inbound/Outbound	No

Tip: When using regional tags, add the tag for the tenant's home region **and** the capacity region (if different), as well as the corresponding paired regions.

1.4.6 Secure Outbound Connectors Matrix

Connection Method	Supported Sources	Fabric Workloads
Trusted Workspace Access	ADLS Gen2 (firewall)	Shortcuts, Pipelines, COPY INTO, Semantic Models
Managed Private Endpoints	Azure SQL, ADLS Gen2, Cosmos DB, Key Vault…	Spark Notebooks, Lakehouses, Spark Jobs, Eventstream
VNet Data Gateway	Azure services in a VNet	Dataflows Gen2, Semantic Models
On-Premises Gateway	SQL Server, Oracle, files, SAP…	Dataflows Gen2, Semantic Models, Pipelines
Service Tags	Azure SQL VM, SQL MI, REST APIs	Pipelines, network integration

1.5 Outbound Protection

1.5.1 Outbound Access Policies

Outbound Access Policies allow you to **restrict outbound connections** from a Fabric workspace to unauthorized destinations.

Objective: Prevent malicious users from exfiltrating data to unapproved destinations.

Characteristics: - **Workspace-level** control - Destinations are allow-listed via **Managed Private Endpoints** or **Data Connections** - Any connection to a destination not explicitly allowed is **blocked**

Availability:

Item Type	Public Preview	GA
Lakehouse, Spark Notebooks, Spark Jobs	Sept 2025	Sept 2025
Dataflows, Pipelines, Copy Jobs, Warehouse, Mirrored DBs	Oct 2025	Nov 2025
Power BI, Databases	Roadmap	Roadmap

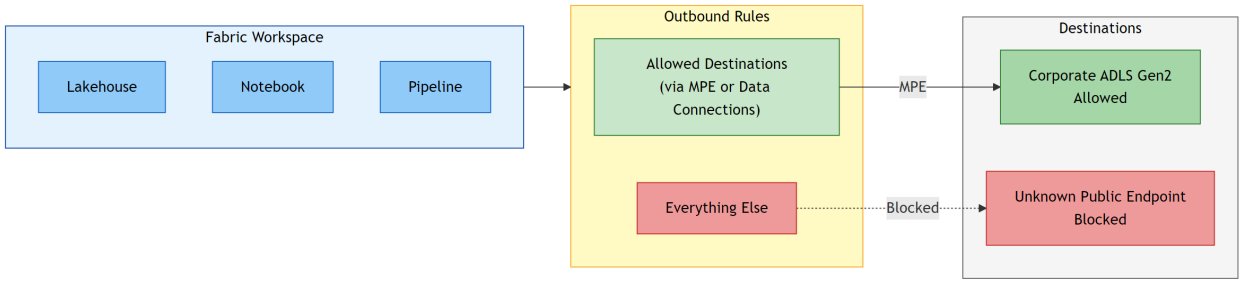


Figure 10: Diagram 10

1.5.2 Data Exfiltration Protection (DEP)

Complete **DEP** is achieved by combining inbound AND outbound protection:

Component	Role
Inbound (Private Link / IP Firewall / Conditional Access)	Controls **who** can access data and **from where**
Outbound (Outbound Access Policies)	Prevents authorized users from **exfiltrating** data to unapproved destinations

1.6 Data Security

1.6.1 Encryption

Type	Mechanism	Detail
In transit	TLS 1.2 / 1.3	All client-to-Fabric and inter-experience traffic
At rest	Microsoft-managed keys (default)	Automatic encryption of all data in OneLake
At rest (advanced)	Customer Managed Keys (CMK)	Second layer of encryption with customer-managed Azure Key Vault keys
Power BI	BYOK	Bring Your Own Key for Power BI datasets

1.6.2 Customer Managed Keys (CMK)

CMK allows organizations to add an **additional encryption layer** with keys they manage themselves in Azure Key Vault.

Availability:

Item Type	Public Preview	GA
Lakehouse, Spark Job, Environment, Pipeline, Dataflow, API for GraphQL, ML Model/Experiment	Aug 2025	Oct 2025
Data Warehouse	Sept 2025	Oct 2025
Databases, Mirrored Experiences, Power BI	Roadmap	Roadmap

1.7 Data Residency and Multi-Geo

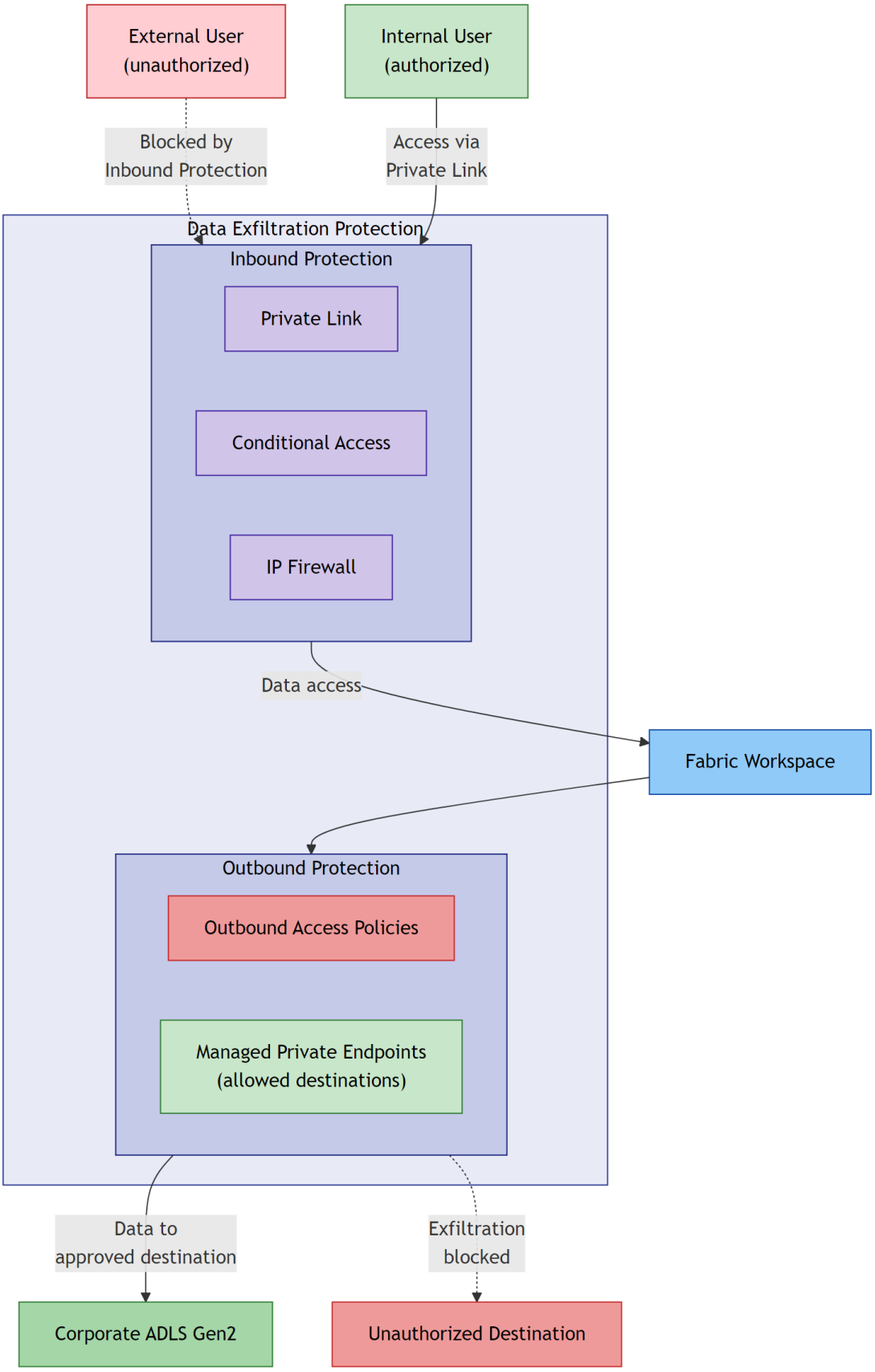


Figure 11: Diagram 11

Some metadata and processing is stored at rest in the tenant's home geography - **Data Residency compliant** by default

1.8 Compliance and Certifications

Microsoft Fabric supports a wide range of compliance standards:

Certification	Date
ISO 27001, 27701, 27017, 27018	December 2023
HIPAA	January 2024
Australian IRAP	February 2024
SOC 1 & 2 Type 2, SOX, CSA STAR	May 2024
HITRUST	September 2024
FedRAMP (Azure Commercial)	November 2024
PCI DSS	January 2025
K-ISMS	May 2025
GDPR, EUDB	Supported

Fabric is a core **Microsoft Online Service**.

1.9 Decision Guide

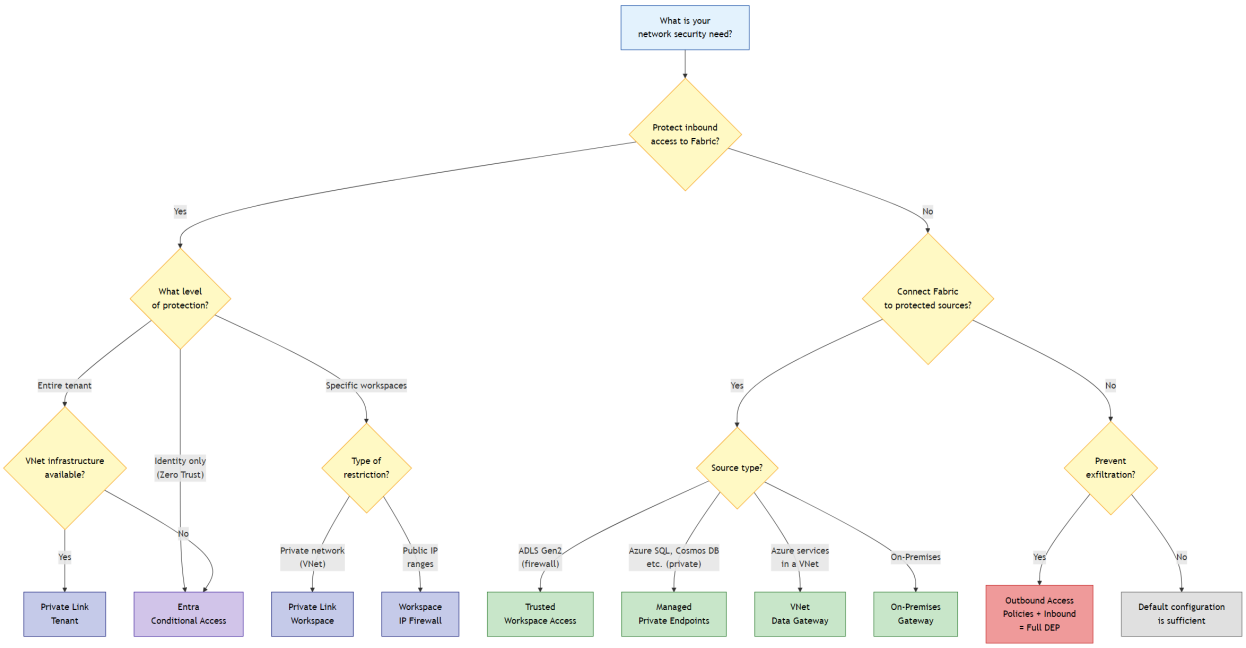


Figure 12: Diagram 12

1.10 Feature Summary and Status

Feature	Level	Direction	Status	Primary Use Case
Entra Conditional Access	Tenant	Inbound	GA	Zero Trust, MFA, IP filtering
Private Link Tenant	Tenant	Inbound	GA	Full tenant network isolation
Private Link Workspace	Workspace	Inbound	GA (Sept 2025)	Granular per-workspace network isolation
Workspace IP Firewall	Workspace	Inbound	GA (Q1 2026)	IP-based restriction without VNet infrastructure
Trusted Workspace Access	Workspace	Outbound	GA	Firewall-enabled ADLS Gen2 access
Managed Private Endpoints	Workspace	Outbound	GA	Private connection to Azure sources
Managed VNets	Workspace	Outbound	GA	Spark isolation + MPE support
VNet Data Gateway	Org	Outbound	GA	Managed connection to Azure services in VNet
On-Premises Gateway	Org	Outbound	GA	Connection to on-premises sources
Service Tags	NSG/Firewall	Both	GA	Azure network rules
Outbound Access Policies	Workspace	Outbound	GA (Sept-Nov 2025)	Data exfiltration prevention
Customer Managed Keys	Workspace	Data	GA (Oct 2025)	Dual-layer encryption

1.11 References

- Security overview - Microsoft Fabric
- Private Links overview
- Workspace-level Private Links
- Managed VNets overview
- Managed Private Endpoints
- Trusted Workspace Access
- Protect Inbound Traffic
- Conditional Access in Fabric
- Service Tags
- Fabric Security Whitepaper
- *End-to-end network security in Microsoft Fabric* – Arthi Ramasubramanian Iyer & Rick Xu, FabCon 2025